

Evaluation ensures recommendations are not just visually appealing, but also statistically meaningful.

Streamlit deployment on Linux

A recommendation engine becomes truly useful only when users can interact with it. I built a lightweight frontend using Streamlit, an open source Python framework. Streamlit runs smoothly on Linux, needs minimal code, has a real-time interactive UI, and is easily deployable on servers.

- The user experience is:
- Enter a book title.
  - Choose model type.
  - Get instant recommendations.
- Streamlit turns ML projects into practical applications.

Deployment architecture

This project can be deployed on open source or cloud infrastructure. Deployment options include:

- Linux VPS server
- AWS EC2 instance
- Elastic Beanstalk
- Docker container

The architecture pipeline is: *Dataset → Cleaning → NLP → Clustering → Recommendation Models → Streamlit UI → Deployment.*

This modular design ensures scalability and maintainability.

You’ve learnt how open source machine learning tools can help to build a hybrid recommendation engine. Such recommendation systems are not only valuable in audiobooks but can also be extended to:

- Movie platforms
- News aggregators
- E-commerce catalogues
- Academic paper recommendations

The best part is, with open source tools, developers have the complete freedom to build, customise, and deploy intelligent systems without proprietary dependencies. **END** 

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| OSFY Magazine Attractions During 2026-27 |                                                                                             |
|------------------------------------------|---------------------------------------------------------------------------------------------|
| Month                                    | Theme                                                                                       |
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| April 2026                               | Open Source Cloud & Infrastructure                                                          |
| May 2026                                 | Blockchain and Open Source                                                                  |
| June 2026                                | Modern Software Development with Open Source   Programming Languages                        |
| July 2026                                | Observability & Performance   Security & Privacy in Open Source                             |
| August 2026                              | All About Data Management (Database management, Optimisation, Big Data, Data analysis, etc) |
| September 2026                           | DevOps, SRE & Platform Engineering                                                          |
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| November 2026                            | Open Source for Edge, IoT & Embedded                                                        |
| December 2026                            | Cloud Special: Everything from Management to Implementation                                 |
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